

Coding Interview Questions (Interview Style)

Arrays

- 1 1. Given an integer array containing numbers from 1 to 100 with one number missing, write a function to find the missing number.
- 2 2. Given an integer array, write a function to find any duplicate number.
- 3 3. Given an unsorted integer array, find the largest and smallest elements.
- 4 4. Given an array of integers and a target sum, return all pairs of numbers that add up to the target.
- 5 5. Given an array containing multiple duplicate elements, return all duplicates.
- 6 6. Given an array, remove duplicate elements and return the result.
- 7 7. Implement the quicksort algorithm to sort an array in-place.
- 8 8. Remove duplicates from an array in-place without using extra space.
- 9 9. Reverse an array in-place.
- 10 10. Remove duplicates from an array without using any library functions.

Linked Lists

- 1 1. Given a singly linked list, find the middle node in one pass.
- 2 2. Determine if a linked list contains a cycle. If yes, find the starting node of the cycle.
- 3 3. Reverse a singly linked list.
- 4 4. Reverse a singly linked list without using recursion.
- 5 5. Remove duplicate nodes from an unsorted linked list.
- 6 6. Find the length of a singly linked list.
- 7 7. Find the third node from the end of a singly linked list.
- 8 8. Given two linked lists representing numbers, return their sum using a stack.

Strings

- 1 1. Given a string, print all duplicate characters.
- 2 2. Check if two strings are anagrams of each other.
- 3 3. Find the first non-repeating character in a string.
- 4 4. Reverse a string using recursion.
- 5 5. Check if a string contains only digits.
- 6 6. Find all duplicate characters in a string.
- 7 7. Count the number of vowels and consonants in a string.
- 8 8. Count the occurrences of a given character in a string.
- 9 9. Generate all permutations of a string.
- 10 10. Reverse the words in a sentence without using library methods.
- 11 11. Check if one string is a rotation of another.
- 12 12. Determine if a string is a palindrome.

Binary Trees

- 1 1. Implement a binary search tree.

- 2 2. Perform a preorder traversal of a binary tree.
- 3 3. Perform preorder traversal without recursion.
- 4 4. Perform an inorder traversal of a binary tree.
- 5 5. Perform inorder traversal without recursion.
- 6 6. Perform a postorder traversal of a binary tree.
- 7 7. Perform postorder traversal without recursion.
- 8 8. Print all leaf nodes of a binary tree.
- 9 9. Count the number of leaf nodes in a binary tree.
- 10 10. Implement binary search on a sorted array.

Sorting & Misc

- 1 1. Implement the bubble sort algorithm.
- 2 2. Implement iterative quicksort.
- 3 3. Implement insertion sort.
- 4 4. Implement merge sort.
- 5 5. Implement bucket sort.
- 6 6. Implement counting sort.
- 7 7. Implement radix sort.
- 8 8. Swap two numbers without using a third variable.
- 9 9. Given two rectangles, determine if they overlap.
- 10 10. Design a vending machine system.